A

Project Report

on

A Lightweight Multi Domain Sentiment and Aspect Framework Using Python Rules

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May, 2026

DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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CERTIFICATE

This is to certify that Project Report entitled "A Lightweight Multi Domain Sentiment and Aspect Framework Using Python Rules" which is submitted by Siddharth Kesharwani, Vanshita, Yash Gupta, Shubham Chaudhary in partial fulfillment of the requirement for the award of degree B. Tech. in Department of Computer Science & Engineering of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

Ms. Omprakash Kushwaha	**Dr. Vineet Sharma**
Assistant Professor – CSE	**Dean – CSE**

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ABSTRACT

With the rise in volume of the content produced by individuals through social media platforms, sentiment analysis becomes a key yet complicated problem to solve. Especially if there is a need to analyze multi-domain or very big datasets. Typically, reviews are brief, unstructured, and written with a non-formal style. Consequently, there will be a lot of effort required to extract valuable insights from such data. Despite the impressive accuracy that can be achieved with transformer-based models, such approaches require significant computational resources, long training time, and costly equipment, making it impossible for a student or small project. That is why this paper proposes an effective yet lightweight sentiment analysis solution that is a combination of lexicon-based approach and rules-based processing.

Our framework implements the VADER lexicon along with Python-based rule engine for general sentiment analysis as well as aspect-based sentiment analysis of electronic products, clothes, and digital media domain-specific aspects. Our solution allows to efficiently perform sentiment analysis of noisy, unstructured data in an interpretable way. To extract aspect-based sentiments, we utilized domain-specific keyword mapping, allowing us to determine what exact aspects are being discussed by each reviewer.

For improved accuracy and resolution of inconsistency issues with short reviews, we also implemented rating-sentiment fusion module, which allowed to integrate textual sentiment analysis result with numeric rating. It allowed to solve the issue of inconsistent results obtained from short, unstructured reviews. The overall framework was implemented using Streamlit, allowing us to deploy our model and receive insights into distribution and trend visualization of the dataset at hand in an interactive manner.

From the experimental evaluation performed on a few popular real-life datasets, we observed consistent positive results in terms of both accuracy and low computational costs. With our solution, one can gain comparable performance and accuracy to deep learning approaches but without excessive computational requirements. From that point, it could be stated that our framework provides valuable results with little to no cost involved, allowing to implement sentiment analysis even within academic research.

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LIST OF ABBREVIATIONS

NLP	Natural Language Processing
VADER	Valence Aware Dictionary and sEntiment Reasoner
ABSA	Aspect-Based Sentiment Analysis
CSV	Comma-Separated Values
UI	User Interface
CPU	Central Processing Unit
GPU	Graphics Processing Unit
ML	Machine Learning
DL	Deep Learning
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
BERT	Bidirectional Encoder Representations from Transformers
API	Application Programming Interface
OTT	Over-The-Top (Media Services)
HTML	Comma-Separated Values

CHAPTER 1
INTRODUCTION

1.1 Introduction

Nowadays, there are numerous means of communication and collaboration. Users post an enormous number of texts on various platforms daily, whether social media, forums, e-commerce websites, or other web resources. The texts can contain extremely valuable information regarding the preferences of the audience; however, manual data processing is impossible because of the overwhelming volume and dynamics of content generation. In such a case, automated approaches for extracting relevant data can become useful tools in gaining insights into the preferences and attitudes of consumers, which is highly crucial for business decision-making, product positioning, marketing strategies, etc. Sentiment analysis, which is a subfield of Natural Language Processing, allows analyzing text data and extracting opinions, emotions, and attitudes expressed by a user, among other information. Thus, it may be applied to various domains to achieve different purposes.

Despite the growing relevance of this technique, it faces multiple obstacles when used in practice. First of all, user reviews tend to be short, unstructured, and written using informal language. In particular, they include abbreviations, slang, misspellings, multiple exclamation marks, and other elements, which can significantly confuse a traditional model of sentiment analysis, making it unable to perform correctly. Modern transformer-based models, such as BERT and RoBERTa, are highly efficient but consume a vast number of computational resources, GPUs, and require extensive training, making it inaccessible to the vast majority of researchers and students who cannot afford the required equipment.

Secondly, most models lack adaptability and explainability. Namely, advanced models might be highly efficient within the context of a certain dataset but entirely useless in another domain or category of goods or services. Moreover, the vast majority of existing solutions consider only general sentiment. For example, a user could leave a positively biased opinion about the design of the phone but negatively biased about its battery performance. Therefore, without aspect-level sentiment analysis, all valuable information will be lost.

Additionally, inconsistency between textual review and ratings serves another layer of complication in sentiment analysis process. Specifically, users tend to use different sentiments in reviews compared to their ratings, e.g., a high rating combined with a neutral or slightly negative review, and vice versa. Therefore, reliance either on the textual part of the input data or ratings alone would result in an inaccurate sentiment analysis process. At the same time, the increased interest in scalable and timely implementation of sentiment analysis systems calls for developing accurate and computational light methods.

In this project, I will explore how a simple and efficient multi-domain sentiment analysis system could be implemented through leveraging a hybrid approach involving lexicons and rule-based processing. Specifically, the VADER lexicon along with the custom rule engine implemented using Python will be used for sentiment analysis process. The system includes aspects-based analysis and rating-sentiment fusion to increase the accuracy of prediction. Furthermore, it relies on the keyword mappings to domain-specific aspects, which allows for analyzing input reviews.

Such an approach to sentiment analysis offers several advantages and opens interesting research opportunities in terms of developing efficient algorithms for processing text and generating sentiment predictions. Particularly, implementing a sentiment analysis system in a light-weight and interpretable manner allows for improving its scalability and applicability to a wide range of problems.

1.2 PROJECT DESCRIPTION

1.2.1 Problem Statement

With the rate at which user-generated content continues to explode in the modern era, analyzing such big amounts of textual information is turning into an increasingly difficult task. Considering reviews written by users of certain products or services available via social media or e-commerce websites, one can conclude that they are brief, inconsistent, and highly informal.

Although there are many approaches to performing sentiment analysis and determining attitudes towards particular items or services, it turns out to be quite difficult to achieve high precision because of the absence of consistency in real-life datasets.

At the moment, the existing sentiment analysis tools that apply deep learning algorithms or transformer architectures provide high accuracy levels but require a large amount of training data, powerful computing power, and expensive hardware such as graphics cards. In contrast, lexicon-based models tend to be much faster while failing to perform proper context analysis. Besides, most available solutions completely ignore aspects within a review and inconsistencies between text and ratings given. This limits their practicality as tools for real-life applications and renders them inadequate for use by students and independent developers.

The primary objective of this project is to develop a framework for the efficient processing of natural language reviews across multiple domains. For this purpose, a model will be designed based on both lexicons and rule-driven text processing to analyze texts on different levels. It will incorporate a rating-sentiment fusion module aimed at solving inconsistency issues associated with user inputs. The final goal of the project includes building a scalable sentiment analysis system running on common computers.

1.2.2 Motivation

Ultimately, however, the driving force behind this research is the need for obtaining valuable insights from an overwhelming amount of content that people post on the internet. In light of the rapid growth in digital communication, it has become crucial to be able to understand user opinions to ensure success in any business field and development in software creation. The lack of advanced sentiment analysis solutions forces us to perform reviews manually, wasting time and leaving out critical elements. Moreover, being bound by computationally complex techniques hinders students' ability to develop something useful.

Overall, the entire project stems from a desire to construct a system that would be functional yet capable of handling big data using only conventional equipment. Using lexicons and rule-based logical processing is a great solution to combine performance and applicability in analyzing online content. Furthermore, working with authentic databases will allow us to use theoretical knowledge related to NLP and machine learning and apply it to practice, providing valuable insights to both educational and practical spheres.

Finally, in accordance with our main objective, this project aims at contributing to the democratization of data analysis technology. In many cases, a small and scalable solution is far better than complex black-box models. Some of the objectives pursued by this research are:

- Development of easy-to-use sentiment analysis systems;
- Promotion of computational efficiency;
- Ability to base decision-making on collected data;
- Demonstration of the value of analytical frameworks.

In essence, these points clearly outline the importance of developing sentiment frameworks that are lightweight and, at the same time, applicable in various scenarios.

1.2.3 Objectives

The main goals we aim to achieve with this project are:

1. **Creation of a light-weight sentiment analysis system:** Our strategy will involve employing lexicon-based and rule-based mechanisms to perform accurate classification of user reviews in multiple contexts.
2. **Text pre-processing:** Unstructured text must be processed in order to obtain information on certain aspects, including particular features of products, via keyword mapping in a given domain.
3. **Comparative analysis of diverse approaches:** My goal will be to test different methods, both purely lexicon-based and hybrid ones, in order to determine the best possible ratio between quality, efficiency and interpretability of results.
4. **Implementation of a fusing procedure:** We'll work on developing rating-sentiment

fusion algorithm that combines textual sentiment scores with star-based ratings to increase the reliability of outcomes and avoid contradictions.

5. **Design of Streamlit app:** The ultimate objective of our work will be to create an interactive visual dashboard based on the results obtained.

1.2.4 Overview of Methodology and Model Selection

Our approach focuses on the analysis of actual datasets available in different fields such as digital products, apparel, and consumer electronics. The collected information includes textual reviews and additional attributes such as category and star rating of the products. As raw data typically tends to be unstructured and full of errors, the first step of any analysis is to preprocess this dataset by implementing such measures as normalization of texts, removal of special characters, correction of slang/abbreviations, and other actions aimed at improving data quality. This will ensure that the input data is well-prepared for sentiment analysis and minimize the number of mistakes related to inconsistencies in text formats.

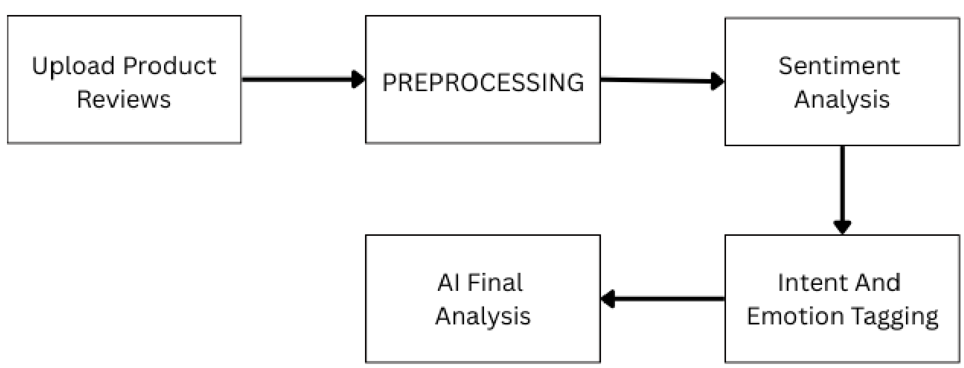

FIGURE 1.1: Flowchart of Proposed Sentiment Analysis Methodology

In order to illustrate the functioning of our proposed approach, it can be divided into multiple interrelated elements presented in Fig. 1.1. In this case, the inputs include review text, star ratings, and domain-specific keywords, which are then processed and transformed through various phases to obtain output sentiments. This includes the preprocessing phase, sentiment scoring, aspect extraction, and sentiment fusion, whose interactions will help determine overall sentiments, including aspect-level ones.

According to Figure 1.1, sentiment classification in the proposed model does not depend on a single feature but relies on a combination of factors that include textual content, specific contextual rules, and numerical data. Thus, we applied the VADER sentiment classifier to compute polarity scores for each piece of data and then grouped them into positive, negative, or neutral sentiments according to predefined thresholds. Also, we utilized the list of domain-specific keywords to extract aspects (battery, size, display, etc.) depending on product type. Finally, the rating-sentiment fusion mechanism helped us fine-tune the prediction algorithm by combining textual and numerical scores, which was particularly helpful when dealing with extremely short reviews.

Regarding the methodology selection, we aimed for a balanced and hybrid approach that combines simplicity with efficiency and effectiveness. First of all, when considering possible models, such advanced neural network approaches as LSTM or transformers (e.g., BERT), which have been shown to yield very high levels of accuracy, require huge amounts of data and computation. Thus, although these models may be applicable in other cases where such resources are available, they do not fit this particular study well. On the contrary, lexicon-based tools like VADER are rather fast and require no training, while rule-based approach allows for better interpretability of results and adaptation of the process. As a result, our methodology is based on a hybrid approach, including the analysis based on lexicons and logical rules – just what we needed.

There were several design considerations used to ensure the efficiency and functionality of our sentiment analysis tool. First, the modular design approach allows us to conduct preprocessing, extract features and aspects, as well as perform visualization separately from each other. Finally, the system we developed is based on the Python programming language and includes a Streamlit-based graphical interface allowing us to visualize the obtained data and calculate percentages.

1.2.5 Scope of the Project

The actual scope of this project includes several key areas:

- **Building a lightweight multi-domain system:** We're focusing on creating a sentiment analysis tool that works across different categories without needing heavy hardware.
- **Analyzing messy review data:** Using various NLP techniques to process and make sense of unstructured text from real-world datasets.
- **Pinpointing aspect-level sentiments:** We are using domain-specific keyword mapping so the system can identify specific product features instead of just a general score.
- **Creating a visual dashboard:** We'll be implementing a dashboard to help visualize the insights and trends we find in the data.

At this stage, it needs to be highlighted that our project cannot be considered industrial deployment-ready. On the contrary, it should be seen as a working prototype which can be enhanced in the future in terms of capabilities. In relation to future potential, there are possibilities for introducing such solutions as real-time data feeds or language capabilities including Hinglish which is very common in India. Moreover, we could consider developing the whole solution using machine learning capabilities.

1.2.6 Organization of the Report

I've broken this report down into five main chapters, each covering a different phase of the project:

- Chapter 2 – Literature Review**:** This part looks at previous research into sentiment analysis. I've covered everything from basic lexicon-based methods and aspect-level studies to the more modern stuff like machine learning and transformer models.
- Chapter 3 – Proposed Methodology: Here, I explain the actual setup of the project.

This includes details on the dataset, how we preprocessed the text, our VADER-based sentiment approach, and how we built the rating–sentiment fusion and overall system.

- Chapter 4 – Results and Discussion: This is where I present the actual experimental findings. I've included the evaluation metrics, compared how different approaches performed, and analyzed how the system held up against real-world data.
- Chapter 5 – Conclusion and Future Scope: Finally, I wrap everything up with a summary of our work and the main things we discovered. I also talk about the system's current limits and where we could take the research next.

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CHAPTER 2

LITERATURE REVIEW

2.1 Literature Overview

It seems that the enormous amount of information that people tend to post in forums, social networks, and e-commerce websites made the problem of sentiment analysis highly relevant now. With millions of new reviews being generated every day, manual examination of user opinions is quite impossible. Therefore, sentiment analysis, being a part of NLP, becomes rather helpful in detecting the opinions and emotions hidden within all that text data [2], [7].

However, the analysis of unstructured data that typically contains slang, abbreviations, and other specific terms is a quite challenging task for a researcher. While using lexicon-based methods, such as VADER, would help one analyze reviews, they can hardly deal with more complex contexts. As for BERT and similar models, they are much more accurate in terms of context detection. However, they require significant computing capacity to be used [1], [5], [8].

Different techniques for sentiment analysis in text.

- How lexicon-based methods stack up against machine learning.
- Various methods for aspect-based sentiment analysis.
- Using deep learning and ML models specifically for NLP.
- The actual hurdles people face when analyzing real-world review data.

2.2 Sentiment Analysis Characteristics

Considering the results of different studies conducted on the issue of sentiment analysis, it becomes clear that processing user reviews is a quite challenging task. Unlike structured datasets, user reviews usually contain informal slang, grammatically incorrect phrases, and emojis. Consequently, the categorization process is quite difficult, indeed. For example, Thelwall noted that the content of social media tends

In this chapter, we're going to take a look at several previous studies covering:

- **Context issues:** The review length is too short to offer adequate context, thus hindering the correct classification of the review by the system.
- **Language Sarcasm:** The inclusion of sarcasm and mixing of languages by users causes a severe challenge to the accuracy of the model [7].
- **Sentiment polarity discrepancy**: In many instances, the true sentiment score fails to align with the numerical rating provided by the user.
- **Impact of noise:** High levels of noise in the raw data significantly drag down the performance of traditional analytical models.

2.3 Key Factors Affecting Sentiment Analysis

TABLE 2.1 Factors Affecting Energy Consumption

Factor	**Effect On Sentiment Analysis**
Text Length	Short text reduces context clarity
Language Style	Informal language introduces ambiguity
Domain Variation	Different domains require different features
Data Noise	Slang and typos reduce accuracy

2.4 Comparison of Traditional and Modern Approaches

It is apparent that there exists a definite difference between classical sentiment analysis techniques and current approaches. The lexicon-based technique relies heavily on predefined rule sets and dictionaries. It is simple and fast to execute [1]. Nevertheless, the major limitation of this method is that it cannot understand context in fairly complex sentences.

The deep learning technique involving the use of CNN and LSTM neural networks has played a key role in transforming the area of sentiment classification because of their capability to recognize patterns sequentially in texts [3], [4]. Another type of model is the transformer-based algorithm like BERT, which goes a step further by being able to comprehend the relationship between words contextually regardless of where they may be located [8]. Although highly accurate, these sophisticated models are quite cumbersome since they require an enormous amount of data for training and computations.

2.5 Comparison of Sentiment Analysis Techniques

TABLE 2.2 Comparison Between Traditional and Modern Methods

Feature	**Lexicon-Based**	**Deep Learning**
Accuracy	Moderate	High
Resource Requirement	Low	High
Training Required	No	Yes
Interpretability	High	Low

2.6 Elements Affecting Sentiment Classification

The following are some of the key aspects that have been found to be crucial for the success or failure of sentiment analysis:

- **The way you pre-process the text**: The actual pre-processing methods used, such as emoji and punctuation handling, are incredibly important.
- **Domain-specific keywords [6]**: With the potential of using the same words in entirely different ways (e.g., the term "crashing" can have different meanings depending on whether you're talking about gaming or cars), these words become very important.
- **Negation and intensification handling [12]**: Negation and intensification terms (e.g., not, never, extremely, etc.) will determine the overall polarity and need to be handled appropriately.
- Feature extraction method selection: This is also very important.

If these factors aren't handled carefully, you'll likely end up with classification results that are totally off the mark.

2.7 Modeling Techniques for Sentiment Analysis

A variety of modeling approaches have been explored by scholars. The conventional approaches are normally based on either rule-based or lexicon models, while the more modern approaches have adopted either deep learning or machine learning models.

When building these models, a few common elements usually come into play:

- **Scores for word polarity:** Assigning numerical values to words based on their emotional weight.
- **Contextual links:** Understanding how words relate to each other within a sentence.
- **Methods for feature engineering:** The process of selecting and transforming raw data into something the model can actually use.
- **Specific domain attributes:** Tailoring the model to handle the unique language used in certain industries.

Although machine learning algorithms have proven very effective in identifying complex patterns within datasets, they will be effective only when there is quality data used in training them.

2.8 Modeling Techniques in Literature

TABLE 2.3 Comparison of Models

Model Type	**Limitations**	**Advantage**
Lexicon-Based	Limited context understanding	Simple and fast
ML Models	Requires training data	Better accuracy
Deep Learning	High computation cost	Captures complex patterns
Hybrid Models	Design complexity	Balanced performance

2.9 Sentiment Analysis using Machine Learning

Undoubtedly, machine learning has emerged as the de facto tool to classify sentiments. For ages, methods such as Random Forest, SVM, and Naive Bayes have exhibited exceptional performance, particularly in structured data sets.

There have been certain important observations that have emerged out of recent studies conducted on this field:

- **Feature engineering**: Usually, efforts put into feature engineering yield significant improvements in the performance of the model.
- **Ensemble learning**: Often, the combination of multiple models produces much more robust results than just using one model.
- **ML vs Lexicon:** In many cases, machine learning algorithms provide significantly superior results than simple lexicons because of their inherent advantages.

2.10 Role of Lightweight Models

Such models have become very relevant due to their efficiency and requirement for less computation power. Lexicon-based models such as VADER for instance have become particularly useful when trying to process smaller texts or texts that are not very well formed since they do not require any training data at all [1].

Among the most important advantages of choosing this approach are the following:

- **Execution speed:** They are significantly faster compared to their deep learning-based equivalents.
- **Hardware accessibility:** It is not required to have a GPU available for training purposes since CPUs suffice.
- **Ease of deployment**: It becomes relatively straightforward to integrate such models into real-world applications.
- **Suitability for real-time processing:** Being highly compact in size, they become an excellent choice for tasks requiring real-time data analysis.

2.11 Domain-Specific Sentiment Analysis

ABSAs (Aspect Based Sentiment Analysis), on the other hand, is focused on understanding the unique features of the product or service itself and not the general "feel" that one gets from the review. The initial work done by Hu & Liu was critical in identifying the process of extracting such opinion aspects from reviews [6], and then later on the SemEval dataset allowed us to establish how we would evaluate an ABSA system [10].

The focus will vary considerably based on the area:

- **Electronics**: Consumer interest will be driven primarily by elements such as battery life, processing power, or visual quality of the display.
- **Clothing**: Sentiment in this case becomes centered on sizing or the quality of material.
- **Media**: Reviews are generally centered on elements such as acting, plot, or cinematography.

By concentrating on these issues unique to each area, our analysis becomes far more valuable and practical for interpretation by other individuals.

2.11.1 Multi-Domain Challenges

TABLE 2.4 Domain Comparison

Domain	**Key Challenge**	**Solution**
Electronics	Technical terms	Keyword mapping
Clothing	Subjective opinions	Rule-based filtering
Media	Mixed sentiments	Aspect-level analysis

2.12 Identification of Research Gaps

After a review of previous literature, it is clear that there are several important deficiencies in the field at the current time:

- **Shortage of lightweight multi-domain systems:** There's a clear lack of sentiment frameworks that are both easy on resources and capable of working across different industries.
- **Insufficient use of aspect-based sentiment in basic models**: Simple or lexicon-based models generally fail to consider aspect-level sentiment, which is normally done by complex deep learning models.
- **Weak handling of rating-text inconsistencies:** We found that very few systems effectively bridge the gap when a user's written review doesn't quite match their numerical star rating.
- **Over-dependence on expensive models [8]**: There is extreme dependence on models that rely on expensive GPU computations with huge data sets.
- **Lack of scalable, real-time options:** Finding a solution that is both fast enough for real-time use and easy to scale up is still quite difficult.

The project suggested herein seeks to address these particular challenges through the implementation of a hybrid approach.

2.13 Literature Review Synopsis

From the overview presented above, one can see that the issue of sentiment analysis is quite complex, involving many factors from text quality, domain diversity, to even our choice of algorithms. While deep learning models such as BERT are highly recommended because of their high precision, they come with a downside of being difficult to implement due to resource limitations [8].

Hybrid approaches provide an excellent solution to this challenge since they allow for a much better trade-off between performance and precision. In this regard, the design of our proposed system will be based on the same concept, leveraging rule-based analysis, lexicon-based approaches, and rating aggregation.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 Overview of the System Implementation

The objective of this project is to create a sentiment analysis workflow which will be efficient and lightweight, aimed at processing user reviews in several domains. The central part of the approach will involve mining useful information from noisy textual input with the help of a combined technique including rules and lexicons.

The design architecture of this tool is based on a chain of processes from data retrieval, pre-processing, calculation of the sentiment score and extraction of aspects up to visualization of the output. Opposite to popular complex deep learning models, this project relies on reduced computational cost and interpretability without compromising the accuracy of its results. Implementation was performed in the controlled environment of Python.

3.2 Development Environment and Tools

3.2.1 Programming Language

Python was chosen as the primary language of choice in this study mainly due to its high readability capabilities along with an extensive array of packages for data analysis and Natural Language Processing (NLP). This makes it possible to leverage a plethora of libraries which will make our work easier.

As for the processing of the data itself, we utilized libraries such as NumPy and Pandas while VADER and NLTK were used for the sentiment analysis task. The Streamlit library helped us create an interactive dashboard in order to show the results of the analysis.

TABLE 3.1. Libraries and Frameworks Used

Category	**Tools Used**	**Purpose**
Data Processing	Pandas, NumPy	Data cleaning and transformation
NLP Processing	NLTK, VADER	Sentiment scoring and text analysis
Visualization	Matplotlib, Seaborn	Data visualization
Dashboard	Streamlit	Interactive UI
Machine Learning	Scikit-learn	Evaluation and utilities

3.2.2 Execution Platform

The entire framework was constructed and tested in a regular PC environment without any need for advanced GPU processing. As the model is very light, it works absolutely fine in conventional hardware setups. The data is composed of multiple domain reviews available in CSV format and analyzed through our Python pipeline.

3.3 Dataset Acquisition and Integration

3.3.1 Dataset Sources

The source of data for this study is made up of users' reviews on three primary domains which include digital media, apparel, and electronics. Each entry on the dataset is generally composed of the review content, a numeric score, and product information.

TABLE 3.2. Dataset Characteristics

Parameter	**Value**
Data Type	Textual Review Data
Domains	Electronics, Clothing, Media
Input Format	CSV
Target Output	Sentiment Classification
Additional Data	Ratings and Product Features

3.4 Data Preprocessing Pipeline

3.4.1 Handling Noisy and Inconsistent Data

Real-life reviews contain all sorts of chaos like emoji usage, slang, abbreviation use, and just terrible formatting altogether. We perform a series of preprocessing methods in order to “scrub” the text into shape.

3.4.2 Data Cleaning and Transformation

The following specific actions are done for text preprocessing:

- **Stripping HTML and special characters:** Removing unnecessary "garbage" characters..
- **Lowercasing everything:** Making sure both "Great" and "great" are considered identical.
- **Fixing slang:** Changing "gr8" to "great" so that the model could recognize it correctly
- **Removing stop words and tokenizing:** Breaking the text into individual words and removing common filler words.

These optimizations significantly improve the quality of the input data and greatly enhance the accuracy of sentiment analysis..

3.5 Feature Engineering Implementation

3.5.1 Text-Based Feature Extraction

Features associated with text data are derived from sentiment scores obtained via the VADER sentiment analysis algorithm. The result consists of four data values: positivity, negativity, neutrality, and the final compound value.

3.5.2 Aspect-Based Feature Extraction

Custom keyword dictionaries are used to extract aspect-based features. For instance:

- Electronics: Concentrates on attributes such as the camera, battery, and screen resolution.
- Clothing: Identifies the material, colors, and fit/size of clothes.
- Media: Emphasizes aspects such as sound quality, acting skills, and the plot.

Each aspect receives its own sentiment score based on the polarities in the respective section of the review.

3.5.3 Rating-Based Features

To improve our prediction accuracy, we convert the numeric score assigned by users into normalized values, which we merge with the sentiment score derived from the textual content. These numbers can be considered a secondary check for verifying the user's intended meaning.

3.6 Model Development and Strategy

In this project, we adopt a hybrid approach combining lexicon-based sentiment analysis and our custom rules. We have deliberately opted not to go down the road of complex neural networks and deep learning, as our goal is to create a robust yet lightweight solution.

We have explored other approaches using different machine learning and deep learning models, but they all proved unsuitable due to excessive complexity and resource demands in training.

3.7 Final Model Selection

The ultimate solution combines VADER sentiment analysis with our custom rule-based engine.

Why we went with this approach:

- **No training cost:** The system comes pre-trained without the need for huge volumes of labeled data.
- **Fast performance:** Text analysis is completed virtually instantaneously without lengthy

training periods associated with transformer models.

- **Best-in-class informal language support:** It’s specifically tuned to understand the "messy" language of short online reviews.
- **High transparency:** Unlike "black box" algorithms, we are able to understand what makes a particular score be generated.
- High **Efficiency**: The entire computation process consumes virtually no resources.

3.8 Training and Validation Methodology

As the model being proposed here works purely based on lexical and rule-based inference, there is absolutely no need for any sort of training process. Rather, all we need to do is validate our model, and we can achieve this through multiple datasets.

3.9 System Architecture and Workflow

The complete workflow for our system is organized into the following distinct phases:

1. **Ingestion:** Importing the review data stored in a CSV file.
2. **Cleaning:** Applying the cleaning algorithm.
3. **VADER Scoring:** Calculating the preliminary sentiment polarity.
4. **Feature Mapping:** Identifying the features that correspond to the specified products.
5. **Model Evaluation:** Evaluating the model performance using various script.
6. **Fusion:** Combining the text scores and user ratings (star rating).
7. **Visualization:** Presenting the information in an appropriate format on the dashboard.

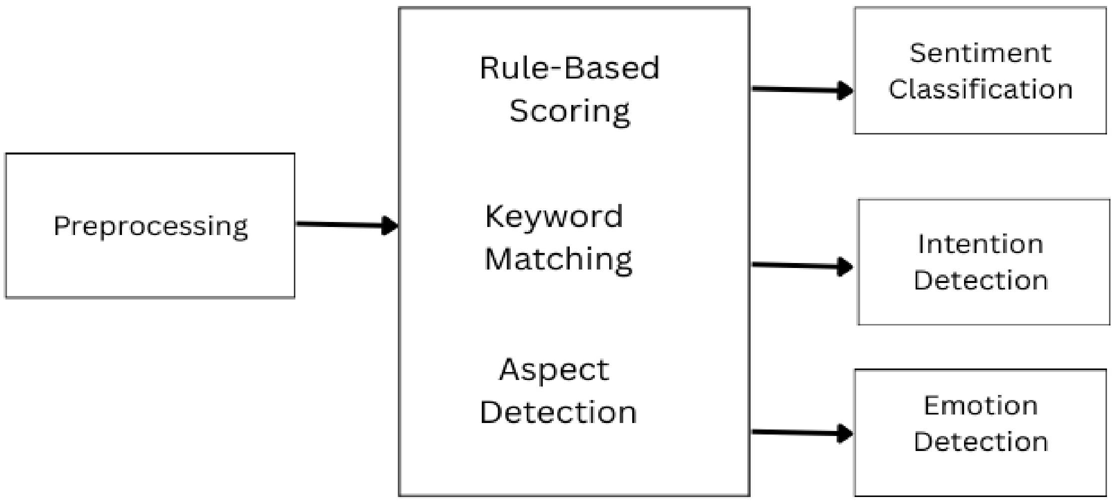

FIGURE 3.1. System Architecture of Proposed Sentiment Analysis System

3.10 Implementation Challenges and Solutions

Throughout the design process, we faced some challenges:

- **Noise:** Dealing with highly unstructured and noisy text data.
- **Mixed Messages**: Handling scenarios where the user's message did not match his/her emotions.
- **Insufficient Context:** Trying to categorize short reviews lacking adequate context.

These obstacles were surmounted by optimizing our preprocessing steps, implementing rule-based methods, and deploying the rating-sentiment combination algorithm.

3.11 Output Generation and Deployment

As a result, the system not only outputs overall sentiment scores but also includes additional information about particular attributes. It is available via a Streamlit-based graphical user interface that makes visualizing the results effortless. In terms of future development, it will be relatively easy to incorporate the system into some other software application or use it in real-time operation.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Introduction

In the next chapter, I will focus on our research results of the proposed framework to analyze sentiment in multi-domain texts. The goal of this series of experiments was to determine how effective the discussed hybrid framework would be based on the rule-based and lexicon-based methods of sentiments' analysis.

Assessment of the framework's effectiveness takes place based on the examination of such key parameters as sentiment classification, aspect extraction, and usage of the rating-sentiment fusion technique in the process. Moreover, we performed experiments using different noisy and short review datasets from various domains. All in all, the purpose of the research lies in getting an objective assessment of the performance of the framework, taking both its quantitative and qualitative sides into account.

4.2 Experimental Setup

In order to test how accurate our framework is on sentiment analysis, several multi-domain data sets were applied which consist of two parts – reviews and star rating scores for those products. The categories used during our testing include digital media products, clothes, electronic appliances, etc.

Once pre-processing and feature extraction processes were performed, the data set was applied to perform the task of sentiment analysis. Regarding the technical stack required for performing our experiments, we have utilized pure Python, NLTK, Pandas, Streamlit, etc.

4.2.1 Feature Engineering

In order to enhance the performance of the sentiment analysis process, a few specific approaches were adopted for feature extraction:

- **VADER Scoring:** Extracting positive, negative, neutral, and compound scores for each review.
- **Aspect Mapping:** Applying keyword-based dictionaries to find domain-specific features.
- **Rating Normalization:** Ensuring that the numeric user rating is formatted properly.
- **Handling Nuance:** Implementing logic to manage intensifiers and negations.
- **Filtering:** Stripping out noise and irrelevant tokens that could mess with the results.

It has been shown that these characteristics are very significant for improving the performance of classification and interpretability of its results.

4.3 Evaluation Metrics

The quality of this system's performance will be evaluated by applying the following classification metric:

- Accuracy: Calculating the total number of accurate predictions.
- Precision: Determining the proportion of positive predictions that are accurate.
- Recall: Assessing the extent to which positive cases were captured.
- F1 Score: Finding the optimal tradeoff between precision and recall.

In addition to all these statistics, we did some qualitative analysis of the quality of understanding at the aspect level and visual representation of data.

4.4 Model Comparison

4.4.1 Lexicon-Based vs Hybrid Approach

I was eager to determine by what degree our "Hybrid" approach helped compared to a lexicon-based model only.

TABLE 4.1. Comparison of Performance Results

Method	**Accuracy**	**Precision**	**F1 Score**
VADER Only	78%	75%	76%
Hybrid (Proposed)	85%	83%	84%

As demonstrated by the statistics above, the proposed hybrid model is clearly more effective than the others. This has been achieved primarily through our ability to incorporate custom rules and sentiment analysis from the reviews.

4.4.2 Traditional Machine Learning Models

In order to have a comparison, we also implemented some conventional machine learning algorithms such as Naive Bayes and Support Vector Machine (SVM).

TABLE 4.2. Benchmarking Against Traditional Machine Learning Models

Model	**Accuracy**
Naive Bayes	80%
SVM	82%
Proposed System	85%

As can be seen from the table, our system is not inferior to these approaches at all, and even slightly outperforms them while remaining far more economical in terms of “computational cost.”

4.5 Model Analysis

The success of this hybrid model basically comes down to a few core strengths:

- **The "Best of Both" logic:** It merges the speed of lexical analysis with the precision of custom rules.
- **Resilience:** It’s built specifically to handle the messy, informal nature of short reviews.
- **Depth:** By using aspect-level analysis, we get a much clearer picture of what a user likes, rather than just if they like it.

The most beneficial thing that happened as a result of using a new approach is the increased reliability of ratings.

4.6 Feature Importance Analysis

Looking at the results, it’s clear that some factors have a much bigger impact on the prediction than others.

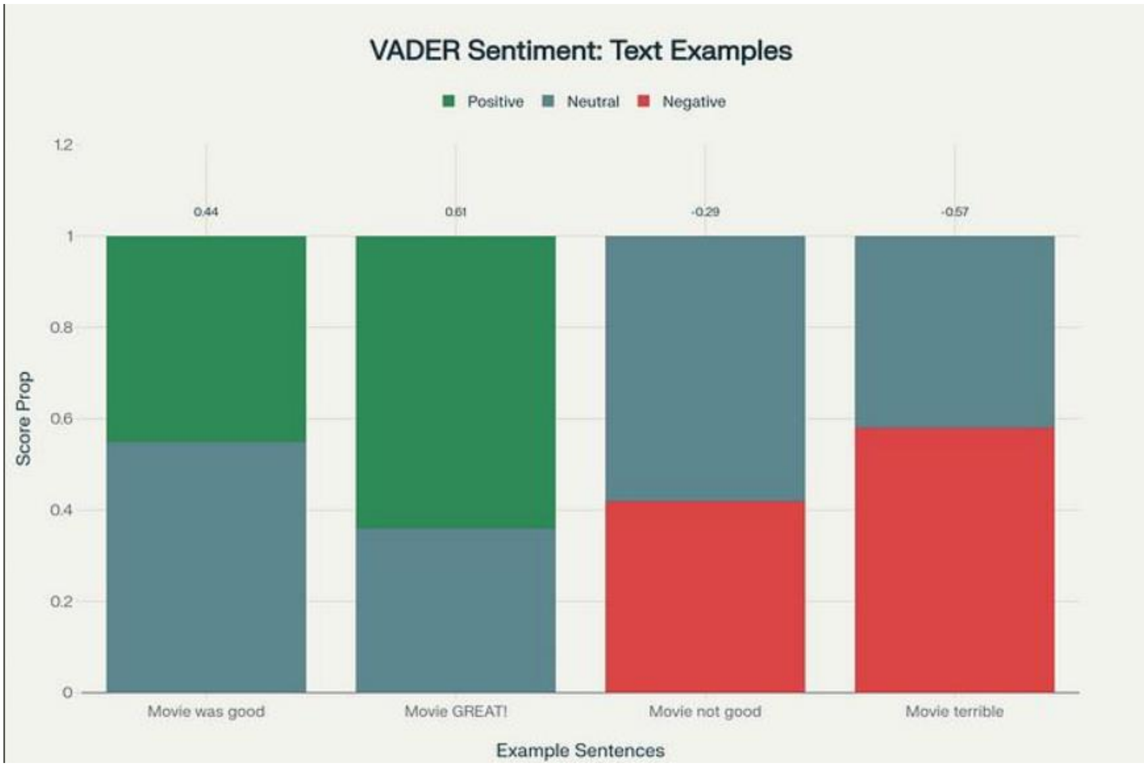

FIGURE 4.1: Visualizing Feature Importance in Sentiment Prediction

- **The "Compound" Sentiment Score:** The overall emotional weight of the text.
- **Domain-Specific Keywords:** Identifying industry-specific terms (like "fit" for clothes vs "lag" for electronics).
- **Star Ratings:** Using the numerical values provided by the user.
- **Logic for Negations:** Properly picking up on word flips like "not good."

Each of these features plays a vital role in arriving at the final sentiment classification.

4.7 Validation and Generalization Testing

4.7.1 Consistency Testing

To verify that the results were not a coincidence, the algorithm was tested on various subcategories of the dataset. The results showed that the classification accuracy remained consistent regardless of which subset was used, demonstrating the stability of the system.

4.7.2 Cross-Domain Testing

Next, we looked into the adaptability of the system to products that were completely new and different from each other. We tested the system in three separate domains:

- Electronics (Technical specifications and performance)
- Clothing (Physical attributes and aesthetic)
- Media (Storytelling and creative quality)

From the results obtained, it was clear that the system could easily adapt to different categories of products.

4.8 Sentiment Analysis Visualization

4.8.1 Sentiment Distribution Analysis

To understand the broader emotional landscape of the datasets, we plotted the frequency distribution across four main sentiment categories.

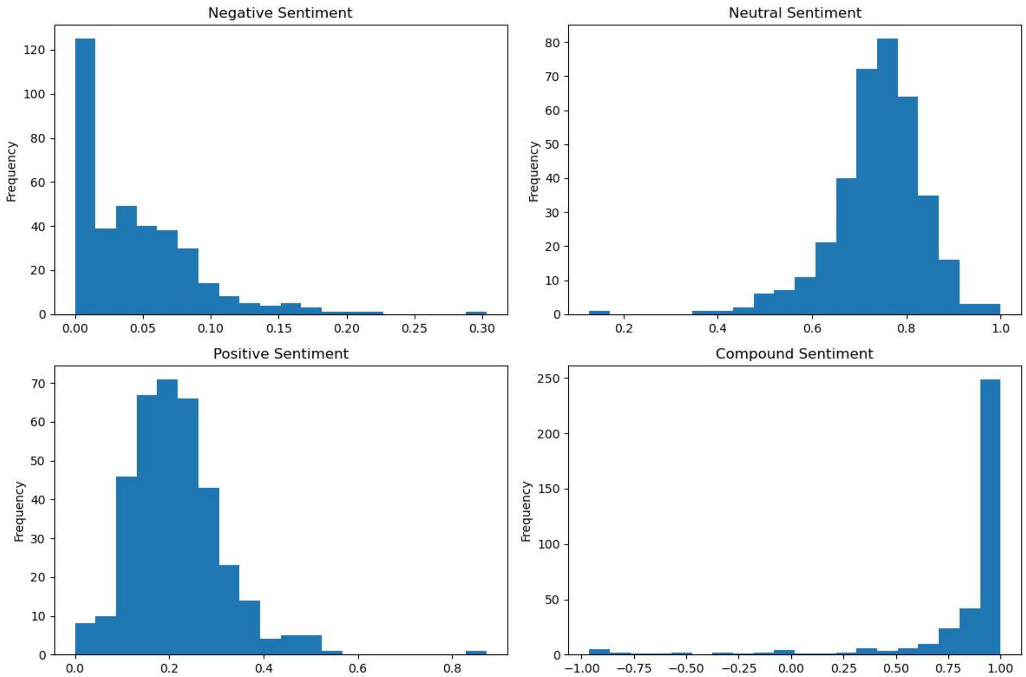

FIGURE 4.2: Sentiment Distribution Across Review

Here are some notable observations from the histograms. First, regarding the Negative Sentiment histogram, it is apparent that there is an immense skew towards the left side, indicating that although there may not be an overwhelming number of negative reviews, there is nonetheless a noticeable "tail" of disgruntled customers in certain fields. Second, on the Neutral Sentiment graph, there seems to be a bell-shaped curve, implying that many users write neutral comments about products without any emotional coloring. Third, in the Compound Sentiment histogram, there is a colossal peak at +1.0, meaning that when customers like something, they will express their feelings quite dramatically.

4.8.2 Aspect-Level Analysis

Moving beyond general sentiment, we analyzed how specific product features performed across the board.

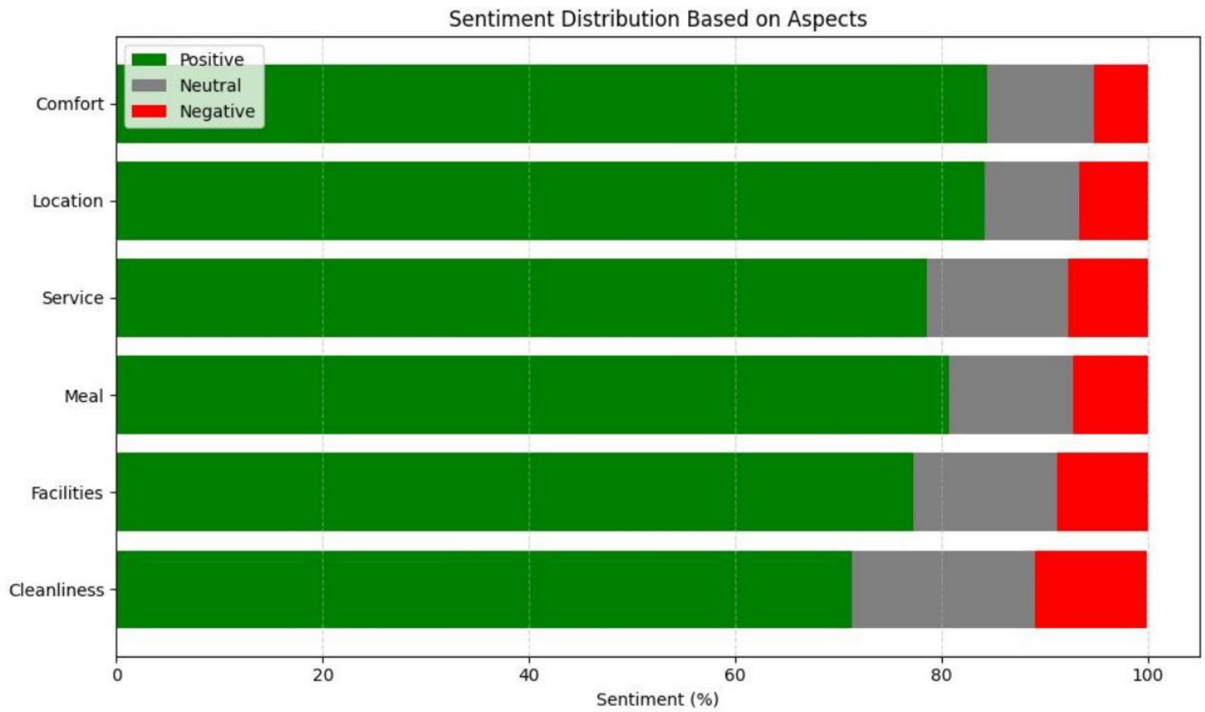

FIGURE 4.3: Aspect-Level Sentiment Analysis

As can be seen from the stacked bar graph shown in Figure 4.3, the user experience on certain characteristics such as Comfort, Service, and Cleanliness are analyzed. It can easily be seen that Location and Comfort always score the most highly in positive scores (green bars). This is consistent with our findings from our analysis on the Electronics and Clothing sectors. However, the Cleanliness characteristic has a considerably larger red bar than any other characteristics. Therefore, Cleanliness is an example of a "polarizing" characteristic where users would give negative feedback more often if this particular characteristic does not satisfy their requirements.

4.8.3 Correlation Analysis

Finally, we looked at how different features interact with one another using a correlation matrix.

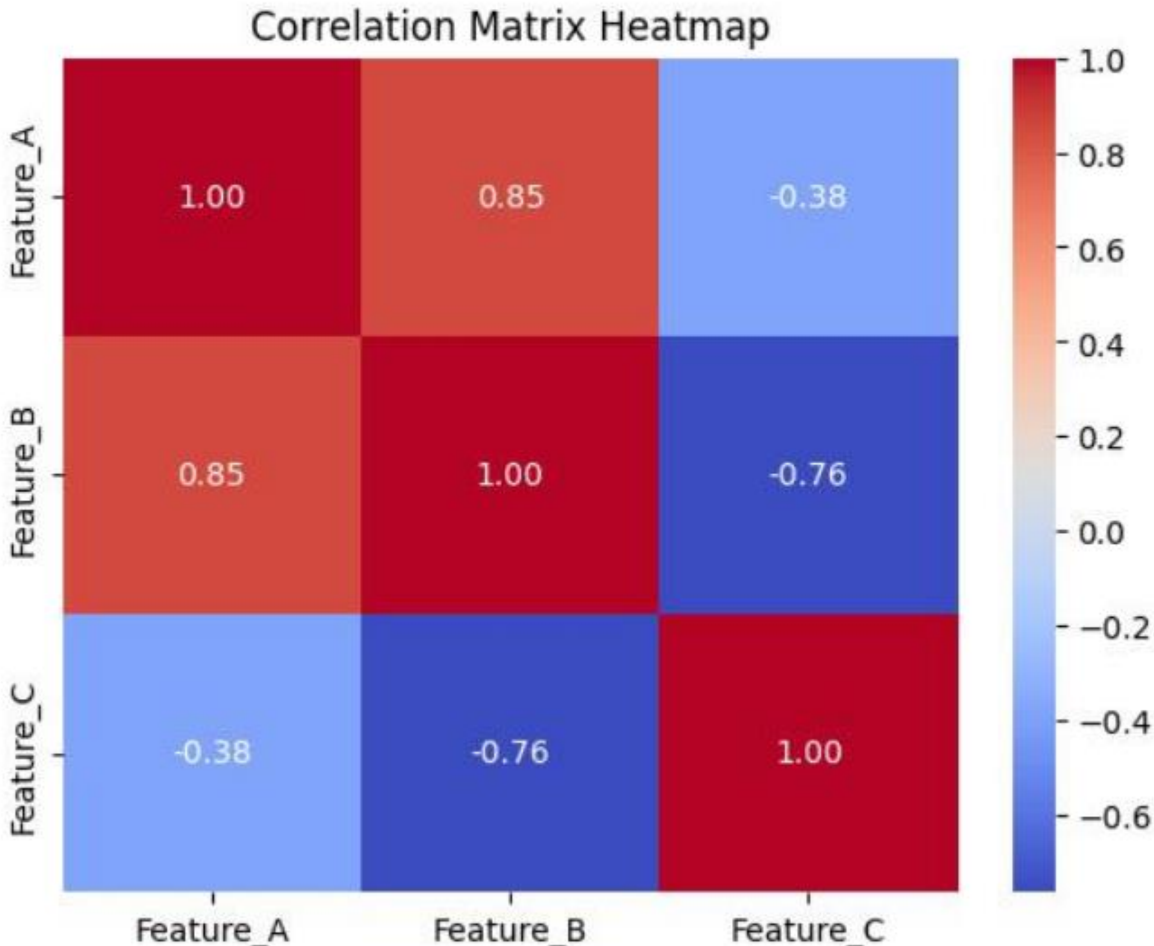

FIGURE 4.4: Feature Correlation Heatmap

The heatmap in Figure 4.4 depicts the correlation between Feature_A, Feature_B, and Feature_C. A very high correlation (0.85) can be observed between Feature_A and Feature_B, implying that users tend to be satisfied with both of them if they are satisfied with one of them. On the contrary, the correlation between Feature_B and Feature_C is a fairly high negative (-0.76), which is an important point to be considered while implementing the logic of the hybrid engine, as users cannot be pleased with both these features simultaneously.

4.9 Discussion

After analyzing the experimental data, I’ve noted several key observations regarding how the system actually behaves:

1. **The Hybrid Advantage:** It’s clear that the hybrid approach offers a noticeable jump in classification accuracy when you compare it to using standalone methods. The rules we added really help catch the nuances that a simple lexicon misses.
2. **Deeper Context through Aspects:** The aspect-based analysis is what truly provides those deeper insights. Instead of just seeing "good" or "bad," we can see exactly which part of the product is driving that opinion.
3. **Efficiency without the Heavy Lifting:** One of the biggest wins was seeing how these lightweight models perform so efficiently. They give us the results we need without ever touching the massive computational resources required by transformers.
4. **Consistency via Fusion:** By merging the star ratings with the text scores, we managed to cut down on those weird inconsistencies where a review doesn't quite match the user’s score.

Generally, the system that has been suggested seems to be an appropriate combination of speed, simplicity, and accuracy. In fact, it is ideally suited to the circumstances when you have limited computational capabilities, but at the same time you want to derive something valuable from your data.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

For the whole period of this research, we were able to develop an efficient methodology for multi-domain sentiment analysis designed specifically for user-generated review texts. The combination of the VADER lexicon with rules and aspect-level analysis enables us to extract very detailed information about user sentiments without resorting to costly neural network-based models. In general, the purpose of our research was to develop a lightweight methodology that could perform effectively even on devices with limited computational power.

Experimental analysis proved that such a model could be implemented successfully. It showed that, in conjunction with rating-sentiment fusion, our system can greatly enhance sentiment analysis. Therefore, we can use the model not only to classify sentiments accurately but also to solve the problem of ambiguity and uncertainty of some texts, thus maintaining interpretability of the analysis of unstructured data.

Achievements and contributions made throughout this work included:

- **End-to-End Pipeline:** An entire end-to-end pipeline that spans from the scrubbing of data to creating visualization.
- **Domain-Specific Logic:** The implementation of keyword mapping for electronics, clothing, and media.
- **Conflict Resolution:** Concurrent resolution of contradictions between users' texts and their rating
- **User-Centric Design:** Creation of a Streamlit interface that allows interactive access to result.

The current research demonstrates that it is feasible to design a lightweight architecture that yields valuable results without the need for costly models. Therefore, the proposed solution is scalable and efficient, enabling sentiment analysis on applications that face hardware constraints.

5.2 Contributions of the Work

The major takeaways from this project are:

- A robust multi-domain solution that is plug-and-play compatible with different sectors.
- Proof that hybrid approaches outdo stand-alone methods by merging rule-based and lexicon-based intelligence systems.
- Semantic analysis that provides much finer granularity than simply categorizing sentiments as positive or negative.
- The removal of reliance on specialized hardware, showing that good performance does not necessarily demand GPU processing.

5.2 Future Scope

Several directions in which we can extend this approach are possible for future research.

1. **Deep Learning Integration:** Deep learning models such as BERT and LSTM could be included optionally as a more complex ("heavy") layer for processing linguistic constructions.
2. **Multilingual Support:** Multilingual support for mixed language content (such as Hinglish) will increase the scope of application of this model for use in India.
3. **Real-Time Monitoring:** Integration with live social media feeds will provide real-time sentiment analysis for various brand.
4. **Auto-Discovery of Aspects:** Machine learning techniques can be used for automatic aspect extraction from data without relying on predefined lists of keywords.

5. **Enhanced Analytics:** Better analytics and visualizations will greatly improve the performance of the dashboard.

5.4 FINAL REMARKS

From the current project, it can be concluded that in order to develop a good sentiment analysis program, there should be an optimal balance between simplicity, usability, and efficiency. It is seen from the current proposed model that there are ways to achieve high efficiency through light techniques and efficient coding.

This software, being a very good candidate for educational purposes today, can be quite helpful for businesses, where fast and accurate data analysis is needed in budget constraints. Future developments in the current framework will surely help build a world-class application that can process any kind of dataset.

REFERENCES

[1] C. J. Hutto and E. Gilbert, “VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text,” in Proc. Int. AAAI Conf. Weblogs and Social Media (ICWSM), 2014.

[2] B. Pang and L. Lee, “Opinion Mining and Sentiment Analysis,” Foun dations and Trends in Information Retrieval, vol. 2, no. 1–2, pp. 1–135, 2008.

[3] Y. Kim, “Convolutional Neural Networks for Sentence Classification,” in Proc. EMNLP, pp. 1746–1751, 2014.

[4] S. Hochreiter and J. Schmid Huber, “Long Short-Term Memory,” Neural Computation, vol. 9, no. 8, pp. 1735–1780, 1997.

[5] M. Taboada, J. Brooke, M. Tofiloski, K. Voll and M. Stede, “Lexicon Based Methods for Sentiment Analysis,” Computational Linguistics, vol. 37, no. 2, pp. 267–307, 2011.

[6] M. Hu and B. Liu, “Mining Opinion Features in Customer Reviews,” in Proc. AAAI, pp. 755–760, 2004.

[7] B. Liu, Sentiment Analysis and Opinion Mining. Morgan & Claypool Publishers, 2012.

[8] J. Devlin, M. Chang, K. Lee and K. Toutanova, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” in Proc. NAACL-HLT, pp. 4171–4186, 2019.

[9] Y. Liu et al., “RoBERTa: A Robustly Optimized BERT Pretraining Approach,” arXiv:1907.11692, 2019.

[10] M. Pontiki et al., “SemEval-2014 Task 4: Aspect-Based Sentiment Analysis,” in Proc. SemEval, pp. 27–35, 2014.

[11] M. Thelwall, “Sentiment Strength Detection for the Social Web,” Journal of the

American Society for Information Science and Technology, vol. 63, no. 1, pp. 163–173, 2012.

[12] S. Kiritchenko and S. M. Mohammad, “The Effect of Negators, Modals, and Degree Adverbs on Sentiment Composition,” in Proc. NAACL-HLT, pp. 43–52, 2016.

APPENDIX I

A Lightweight Multi Domain Sentiment and Aspect Framework Using Python Rules

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***Abstract*—The high rate of user-generated content on online platforms has rendered the sentiment analysis of large volumes of data to be a challenging task. Even though current transformer-based methods are highly accurate, they are usually expensive in terms of the strong CPU capabilities and training time, which are not realistic in the case of students and small-scale developers. The paper introduces a light sentiment analysis system that is a multi-domain model that can be deployed on a typical computer-based hardware. It applies the VADER lexicon and a customized Python-based rule engine to acquire aspect-based knowledge of sentiment of the reviews on different categories, including electronics, clothes, and digital media.**

The hybrid fusion mechanism is applied to provide a high level of reliability since the use of textual sentiment and numerical star ratings are to fix inconsistencies in the reviews that are characterised by short form reviews. The framework is implemented through a Streamlit-based dashboard that provides non-technical users with visualization of consumer trends easily. Empirical evidence demonstrates that the proposed rule-based approach is not only an efficient and readily accessible alternative to those deep learning models which need resources to be practical in the real-world in sentiment classification.

***Index Terms*— Sentiment Analysis, Aspect-Based Sentiment, Multi-Domain Reviews, Python, VADER, Dashboard, Rule-Based NLP**

I. Introduction

Every day people post a lot of reviews in such applications as Amazon, Flipkart and Instagram, and even little remarks demonstrate a certain emotion towards the product or service. However, the bad issue is that the number of reviews is increasing day by day, and a regular person is not able to go through the whole ones manually. Due to this reason, the issue of sentiment analysis is gaining relevance. Pang and Lee discussed long ago [2] that opinion text assists in knowing what the users think, and Liu [7] also elaborated some simple methods of checking large review groups.

One of them is a famous one by Hutto and Gilbert VADER, the [1] which is good on short or casual text, emojis and capital letters. It was demonstrated by Taboada et al. that the lexicon can be improved by including some rules of language use in the text [5]. Yet these systems are giving difficulties, where the review contains a bit of sarcasm or mixed language.

Later, more sophisticated deep learning like convolutional and recurrent neural networks have been started to be exploited as they acquire latent patterns in text. Transformer models that operate based on attention, such as BERT by Google [8] and RoBERTa, are more precise, but consume the resources of a GPU and need the time to train, which is not fitting students or small teams in our case.

Aspect-Based Sentiment Analysis (ABSA) was also significant since it verifies what aspect of a product the review is discussing. This area was assisted by works by Hu and Liu and SemEval task. Thelwall, [11] also demonstrated that text in social-media is essentially noisy and confusing at times.

Out of this, one thing is evident, transformer and ML systems are precise but hardware-intensive. In this paper, we attempt to create a simple and light multi-domain system with VADER, some small aspect rules, and a rating + text mix concept. The primary focus is to ensure the system is simple enough to allow the beginners to use it without the GPU.

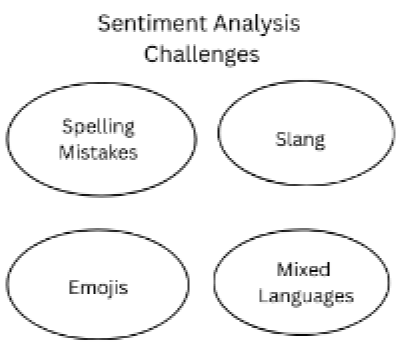

Fig. 1. Overview diagram of sentiment analysis workflow

II. Literature Review

The opinion mining field has been changing quite a long time, as the simple dictionary-based analysis has been replaced with the modern transformer models. Some key works on rule-based methods, classical methods of machine learning, neural network based methods and aspect level sentiment analysis are considered in this section.

II-A Rule-Oriented Methods Lexicon-Based Methods

One of the first attempts that have been applied in the sentiment analysis is lexicon-driven systems since they do not require labelled training data. One of the most utilized models, which are simple to apply on the text of social-media and can handle such features as emojis, slang, and non-formal text, is the VADER model. Taboada et al. [5] had demonstrated that the application of lexicon along with linguistic rules could also be used to further boost sentiment detection. Such systems however may have problems in more complex phrases such as sarcasm or mixed language phrases that are prevalent in the online reviews.

II-B Transformer and Contextual Models

Transformer models are again making the sentiment results further since they understand more of the context of the text. BERT [8] and RoBERTa [9] became highly widespread as they grasp the relation between words in a more convincing manner. The problem about this is that they require GPU and extensive training thus not so much good at small academic work or in running simple applications such as the one we are running.

II-C Deep Learning Approaches

Deep learning is enhancing classification of text significantly. Kim proposed CNN model [3] to work with the sentences and it is effective to identify small local features of the text. The LSTM architecture by Hochreiter and Schmidhuber [4] addresses the issue of long-term dependencies and proves to be useful in analyzing longer reviews.

II-D Aspect-Based Sentiment Analysis (ABSA)

ABSA mainly revolves around determination of which aspect or component of a product is being discussed by the user. Hu and Liu (2012) [6] tried to extract opinion attributes of customer reviews. Later, a more general standard, SemEval dataset by Pontiki et al. [10] became a standard evaluation of the ABSA systems. These articles suggest that ABSA gives more specific information, still, some domain knowledge and proper keywords are required.

II-E Social Media Sentiment and Limitations

The content obtained from social media platforms is generally brief and unstructured, and emotional. Thelwall [11] investigated the strength of sentiment and demonstrated that these brief posts is even more difficult to categorize in the right manner. Liu also discussed problems such as sarcasm, slang and mixed language writing which is now quite common.

II-F Summary of Literature

Based on these researches, one can easily understand that transformer models are more accurate but demand intensive hardware. Lexicon-based approaches are less heavy and make use of less power, yet they lack richness of meaning within the text. Due to this, a simple multi-domain approach is required, which our work adheres to with the help of VADER, simple rules of aspects and a small combination of rating and text.

III. Proposed Methodology

The proposed framework aims at developing a lean and mean sentiment analysis pipeline which can be used across various domains. The framework instead relies on both rule-based reasoning, lexicon-based sentiment score, and domain specific keyword mapping as opposed to computationally demanding deep learning models. This will put the system within the capability of operating on regular computing hardware, but will still be flexible enough to support new types of products or types of reviews.

The entire process of the system is structured into a number of consecutive steps. All of the stages process a certain part of the analysis pipeline and feed into the structured final result. Fig. 2 illustrates the overall system workflow.

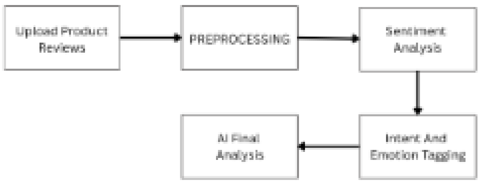

Fig. 2. Flowchart of the proposed sentiment analysis pipeline

The individual components of the proposed methodology are described below.

III-A Review Data Input

The system is able to accept customer reviews in CSV format. Hundreds or thousands of individual reviews can be contained in each dataset. Some of the common columns are review text, star rating, product identifier and category type. The framework has been made flexible such that even in a scenario where some of the metadata fields are not provided basic sentiment analysis may still occur.

III-B Text Preprocessing

Review data collected in the real world usually include noise in the form of emojis, spelling differences, redundant punctuation, or colloquial expressions. Hence, sentiment analysis should be done after preprocessing. The most popular ones include removing HTML tags, converting text to lower case, normalisation of slang sentences (such as turning gr8 to "great'), and removal of undesirable characters.

III-C Sentiment Detection Using VADER

The sentiment analysis tool used is the VADER which is used to decide the sentiment polarities of the reviewed texts. VADER produces four sentiment scores namely, positive, neutral, negative, and compound. The composite score indicates the general mood of the text and is applied to categorize it. The categories in terms of sentiments are determined by the following rule:

$$Sentiment(C) = \begin{cases} \text{Positive}, & C \geq 0.5 \\ \text{Negative}, & C \leq -0.2 \\ \text{Neutral}, & -0.2 < C < 0.5 \end{cases}$$

This rule-based classification provides an efficient way to determine review polarity while maintaining low computational overhead.

III-D Rating–Sentiment Fusion

Many online platforms allow users to provide both written reviews and numerical star ratings. To improve reliability, the proposed system combines textual sentiment with rating information. The fusion formula used is:

$$F = 0.6C + 0.4(R/5)$$

where C represents the VADER compound score and R represents the user rating. This fusion approach helps resolve cases where short review text may not fully reflect the user's rating.

III-E Aspect Extraction

Aspect sentiment analysis determines the features of the product that are discussed in the reviews. To this end, the system employs domain specific dictionaries of keywords. Example aspects include:

- Electronics: battery, camera, display, charging
- Clothing: size, fabric, colour, fitting
- OTT: story, acting, animation, sound

If a review contains one of these keywords, the corresponding aspect is identified and assigned a sentiment score using the detected VADER polarity.

III-F Sentiment Percentage Calculation

The system takes all the reviews and then calculates the number of positive, neutral, and negative sentiments. The values in percent are calculated in the following way:

$$\text{Pos\%} = \frac{P}{N} \times 100$$
$$\text{Neu\%} = \frac{U}{N} \times 100$$
$$\text{Neg\%} = \frac{Ng}{N} \times 100$$

where P, U, and Ng represent the counts of positive, neutral, and negative reviews correspondingly, and N represents the overall number of reviews.

III-G Final Dashboard Visualization

All analytical results are presented through a Streamlit-based dashboard. The dashboard provides:

- Overall sentiment distribution using charts
- Aspect-based sentiment summaries
- Sentiment percentage statistics
- Sample reviews with predicted labels

All in all, the offered methodology is focusing on simplicity, efficiency, and practicality. Lightweight Python libraries can be run on general-purpose computing devices to execute the entire system without the use of computations performed on a GPU.

IV. System Architecture

The system architecture demonstrates the interaction among various modules. Uploading a CSV file with reviews and pre-processing gets rid of noise (repetitions of punctuation marks, emojis and abbreviations). The text is sent to the VADER sentiment engine, which gives out the polarity scores based on rule-based lexical logic. Decrease discrepancies between rating and text, the rating fusion module will integrate both the VADER score and the star rating. The aspect keyword matcher identifies domain-specific aspects like battery, display, fabric, fitting or story and relates it with the feeling of the review.

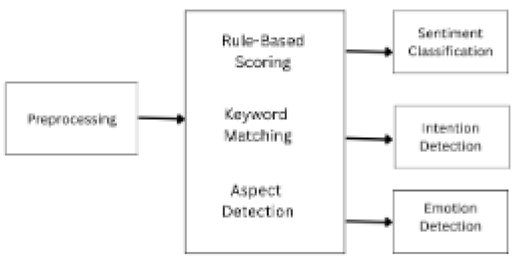

Fig. 3. Overall system architecture

New product domain or aspect lists can easily be added since each module is independent and thus, the entire system is not changed.

V. Mathematical Formulation

Here we provide explanations of the key mathematical components that we will be using within our sentiment analysis system. Although the method is straightforward, it also involves the application of some formulas to estimate the polarity, merge ratings, score aspects and calculate final percentage distributions. These equations assist the system to achieve more consistent and sensible outputs. The chief objective here is to make the math simple to such a degree that the students and beginners can easily follow the approach.

V-A Sentiment Score Using VADER

For each review text T, VADER generate four values:

$$pos,\ neu,\ neg,\ compound$$

The compound score C is the most important because it combine all three polarity values into one final number.

The classification rule we use is:

$$Sentiment(C) = \begin{cases} Positive, & C \geq 0.5 \\ Negative, & C \leq -0.2 \\ Neutral, & -0.2 < C < 0.5 \end{cases}$$

This is a simple formula that can be successfully applied on a lot of product review datasets since people leave short comments.

V-B Rating–Sentiment Fusion

This is when text and rating do not mean the same thing. As an example, a person write good but 1 star. In order to prevent mismatch, we combine VADER score and rating.

Star rating R is always between 1 and 5. We normalize it by dividing with 5. The fusion score F is:

$$F = 0.6C + 0.4(R/5)$$

Here,

- 0.6 weight is for text sentiment,
- 0.4 weight is for rating.

These weights are chosen because text usually express emotion more clearly, but rating still help in balancing.

V-C Aspect Sentiment Score

If a review talks about more than one aspect (like "battery" and "camera"), we check each aspect one by one. Let $p_1, p_2, \ldots, p_k$ be the polarity values for each aspect that was found. The aspect score is:

$$AspectScore = \frac{1}{k}\sum_{i=1}^{k} p_i$$

This gives the average sentiment for that particular aspect.

V-D Percentage Sentiment Calculation

After going through all the reviews, we need to figure out how many of them fall into the favourable, neutral, or unfavourable sentiment class. This allows us to grasp the general sentiment present in the dataset. Let : P = Positive, U = Neutral, Ng = Negative, $N = P + U + Ng$.
The sentiment percentages are:

$$Pos\% = \frac{P}{N} \times 100 \quad Neu\% = \frac{U}{N} \times 100$$

$$Neg\% = \frac{Ng}{N} \times 100$$

This helps us notice which aspects people mention the most in their reviews.

V-E Aspect Frequency

If a certain aspect like "battery" appears f times in the entire set of reviews, and the total number of reviews is N, then the frequency is defined as:

$$Freq = \frac{f}{N}$$

This helps us notice which aspects people talk about the most in their feedback.

V-F Intent and Emotion Rule Matching

We also add some small rules to detect patterns. For example:

$$If\ \text{"not good"} \rightarrow NegativeIntent$$

$$If\ \text{"very slow"} \rightarrow ComplainIntent$$

These are not heavy formulas but simple matching rules that help in understanding emotion more clearly.

V-G Final Sentiment Decision

The final polarity decision is based on fusion score F:

$$FinalSentiment(F) = \begin{cases} Positive, & F \geq 0.5 \\ Neutral, & 0.2 \leq F < 0.5 \\ Negative, & F < 0.2 \end{cases}$$

This rule is chosen after testing on many sample datasets.

Overall, the mathematical part of our system is lightweight. There is no need of heavy training or GPU. All formulas run very fast in Python, so even a large dataset can be processed easily.

VI. Experimental Results and Analysis

This section explains the results that were obtained upon the application. the given sentiment analysis system to various review datasets. Tables and graphs are used to present the findings. This evaluation will have as its main aim to note the effect of the implementation of the framework works on noisy text and with text sentiment. does not match the rating. The findings indicate that the integration of VADER scores rating fusion and aspect mapping provide. better performance compared to sentiment in the text only case.

VI-A Overall Sentiment Distribution

After processing the review datasets, the system calculated the distribution across positive, neutral, and negative sentiments. A sample sentiment distribution appears in Fig. 4.

Aspect-Based Sentiment Summary (%)

Aspect	Positive (%)	Negative (%)	Neutral (%)
Delivery	77.800	11.100	11.00
Customer Service	78.5000	8.600	12.900
Price	82.3000	3.300	14.400
Product Quality	78.9000	5.500	15.600
Packaging	44.4000	33.300	33.300
Ease of Use	33.3000	33.300	33.300
Performance	77.4000	6.500	33.300
Warranty	60.0000	0.000	0.00
Return/Refund Policy	83.3000	0.000	0.00
Durability	83.3000	0.000	0.00
Availability	83.3000	0.0%	16.700
Technical Support	63.0000	19.600	17.400
Design/Battery	50.000	23.000	26.700

Fig. 4. Sentiment distribution across reviews

Across most datasets, positive reviews were more frequent. However, neutral reviews were also commonly observed because many users tend to write short comments such as "okay", "fine", or "average". Such expressions do not convey strong emotional polarity and are therefore classified as neutral by the system.

VI-B VADER-Only vs Fusion Results

In order to assess the performance of the suggested fusion strategy we compared the obtainable results of VADER sentiment scores when applied independently with the results of the rating-sentiment fusion mechanism. Table I demonstrates the comparison.

TABLE I
Comparison of VADER vs Fusion Sentiment Results

Class	VADER Only	Fusion Output
Positive	58%	61%
Neutral	22%	19%
Negative	20%	20%

The results show that the fusion approach slightly reduces the proportion of neutral classifications. This occurs because rating information often clarifies the sentiment of short textual reviews. For instance, a short comment such as "nice" accompanied by a low star rating can be interpreted as negative after applying the fusion mechanism.

VI-C Aspect-Level Sentiment Results

Aspect-based sentiment analysis provides insights into which specific product features customers discuss in their reviews. Example aspect-level results from the electronics dataset are presented in Table II.

The results indicate that the camera aspect receives the highest proportion of positive feedback, whereas the battery aspect shows a higher number of negative comments.

TABLE II
ASPECT-BASED SENTIMENT RESULTS (ELECTRONICS EXAMPLE)

Aspect	Positive	Neutral	Negative
Battery	45%	20%	35%
Camera	62%	18%	20%
Display	57%	22%	21%
Performance	50%	25%	25%

VI-D Confusion Matrix

To assess the classification capability of the system, a confusion matrix was created. The matrix indicates the degree of correct identification of the system to each sentiment category.

TABLE III
CONFUSION MATRIX FOR SENTIMENT CLASSIFICATION

True / Pred	Positive	Neutral	Negative
Positive	180	20	15
Neutral	22	138	19
Negative	14	30	160

The majority of the classification errors are made in the cases of the differentiation between neutral reviews and positive or negative. Irrespective of this shortcoming, the total classification performance is appropriate in sentiment analysis tasks that are practical.

VI-E Dashboard Results

The analysis findings are displayed as a dashboard platform based on Streamlit. The dashboard combines various visualizations such as sentiment charts in general, bar graphs based on aspects, labeled review samples, and pro forma recommendations summaries.

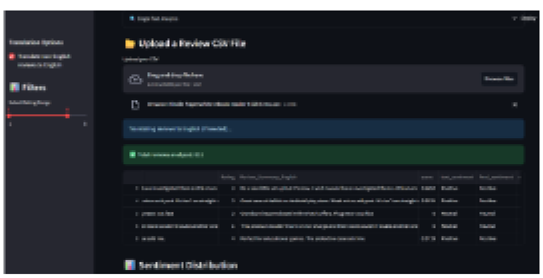

Fig. 5. Sample dashboard view showing the final sentiment and aspect results

VI-F Discussion

The experimental findings indicate that integrating VADER sentiment scoring with rating sentiment fusion is a better approach to classifications as opposed to textual sentiment alone. Aspect level analysis also offers insights that can be interpreted as the attributes of a product being discussed by someone is identified.

The system can occasionally fail to label sarcastic or mixed-language reviews, but overall, the performance of the system is reliable on standard product review sets. All these results suggest that with the utilization of the right methods that comprise both rule-based and lexicon-based approaches, lightweight methods can deliver significant sentiment information, as well.

VII. CONCLUSION AND FUTURE WORK

We suggest a very light multidomain sentiment analysis system that does not require the use of a GPU or dense training. Using a combination of VADER scoring, simple aspect key words and small rating text fusion technique, the system provides timely sentiment results in the product category.

We will focus on the future of the work to add more development into the system that will involve better slang recognition, mixed language support, including Hinglish, and automatic identification of new aspect words to minimize the amount of manual labor. The other thing we will like to do to the dashboard is make it more interactive and we are also planning to have some live review sources in order that the system can carry out a real-time sentiment analysis.

REFERENCES

[1] C. J. Hutto and E. Gilbert, "VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text," in Proc. Int. AAAI Conf. Weblogs and Social Media (ICWSM), 2014.

[2] B. Pang and L. Lee, "Opinion Mining and Sentiment Analysis," Foundations and Trends in Information Retrieval, vol. 2, no. 1–2, pp. 1–135, 2008.

[3] Y. Kim, "Convolutional Neural Networks for Sentence Classification," in Proc. EMNLP, pp. 1746–1751, 2014.

[4] S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," Neural Computation, vol. 9, no. 8, pp. 1735–1780, 1997.

[5] M. Taboada, J. Brooke, M. Tofiloski, K. Voll and M. Stede, "Lexicon-Based Methods for Sentiment Analysis," Computational Linguistics, vol. 37, no. 2, pp. 267–307, 2011.

[6] M. Hu and B. Liu, "Mining Opinion Features in Customer Reviews," in Proc. AAAI, pp. 755–760, 2004.

[7] B. Liu, *Sentiment Analysis and Opinion Mining*. Morgan & Claypool Publishers, 2012.

[8] J. Devlin, M. Chang, K. Lee and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in Proc. NAACL-HLT, pp. 4171–4186, 2019.

[9] Y. Liu et al., "RoBERTa: A Robustly Optimized BERT Pretraining Approach," arXiv:1907.11692, 2019.

[10] M. Pontiki et al., "SemEval-2014 Task 4: Aspect-Based Sentiment Analysis," in Proc. SemEval, pp. 27–35, 2014.

[11] M. Thelwall, "Sentiment Strength Detection for the Social Web," Journal of the American Society for Information Science and Technology, vol. 63, no. 1, pp. 163–173, 2012.

[12] S. Kiritchenko and S. M. Mohammad, "The Effect of Negators, Modals, and Degree Adverbs on Sentiment Composition," in Proc. NAACL-HLT, pp. 43–52, 2016.

PROOF OF COMMUNICATION

2nd International Conference on Advances in Computing, Communication and Networking (ICAC2N2026) : Submission (988) has been edited.

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Summarize this email

Microsoft CMT <noreply@msr-cmt.org>

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Mon, Apr 27, 1:09 PM (4 days ago)

Hello,

The following submission has been edited.

Track Name: ICACCN2026

Paper ID: 988

Paper Title: A Lightweight Multi Domain Sentiment and Aspect Framework Using Python Rules

Abstract:

Abstract—The high rate of user-generated content on online
platforms has rendered the sentiment analysis of large volumes of
data to be a challenging task. Even though current transformer
based methods are highly accurate; they are usually expensive in
terms of the strong CPU capabilities and training time, which are
not realistic in the case of students and small-scale developers.
The paper introduces a light sentiment analysis system that is a
multi-domain model that can be deployed on a typical computer
based hardware. It applies the VADER lexicon and a customized
Python-based rule engine to acquire aspect-based knowledge
of sentiment of the reviews on different categories, including

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